225797: porphyritic microgranite, Three Peak Hill

(Intrusion in Duck Creek Dolomite, Wyloo Group, Ashburton Basin, Capricorn Orogen)

Location and sampling

Yarraloola (SF 50-6), Pannawonica (2154)
MGA Zone 50, 402247E 7598526N

Sampled on 24 July 2018

This sample was collected from outcrop on the north side of the Rio Tinto railway access track on Yarraloola Station, about 5.3 km northeast of Observation Hill, 1.7 km southwest of White Gate Well, and 1.0 km northwest of Three Peak Hill.

Geological context

The unit sampled is a porphyritic microgranite with sill-like geometry intruded at the contact between the 2008–1799 Ma Duck Creek Dolomite and the overlying Ashburton Formation (Williams et al., 1968; Thorne and Seymour, 1991; Thorne and Johnson, 2020a,b). An igneous crystallization age of 1798 ± 9 Ma was determined for a dyke of broadly similar composition intruded into the upper Fortescue and lower Hamersley Groups about 17.9 km to the east-northeast (GSWA 239798, Wingate et al., 2021). Dacitic rocks about 62 km to the south dated at 1790–1787 Ma were tentatively assigned to the Capricorn Group (Evans et al., 2003; GSWA 219588, Wingate et al., 2019). Intrusive rocks of similar age farther south include granitic rocks of the 1820–1775 Ma Moorarie Supersuite of the Gascoyne Province (Sheppard et al., 2020).

Petrographic description

The sample is a porphyritic microgranite (Fig. 1) of dacitic composition, consisting of about 55% groundmass, 30–33% plagioclase phenocrysts, 5% ex-ferromagnesian phenocrysts, 3% magnetite, and minor chlorite, epidote, apatite, and pyrite. The groundmass is massive and composed of a microcrystalline quartz–feldspar aggregate with a poorly defined micropoikilitic texture. Plagioclase phenocrysts are strongly clay altered and all ferromagnesian phenocrysts have been replaced by bright green chlorite and minor epidote. Plagioclase phenocrysts are subhedral to euhedral, tabular, and up to 4.5 mm long. The ex-ferromagnesian phenocrysts are subhedral to euhedral, tabular, and up to 2 mm long; some have forms suggestive of amphibole or pyroxene. Magnetite forms subhedral embayed grains up to 1.5 mm in size, and have been partially altered to extremely fine titanium minerals (titanite and/or rutile). Apatite forms euhedral laths up to 0.5 mm in size. Rare disseminated grains of anhedral to subhedral, fractured pyrite are up to 0.45 mm in size. The sample is crosscut by subparallel epidote–carbonate veins up to 0.7 mm wide. The rock is not deformed and not obviously metamorphosed.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown, and mostly subhedral to euhedral. The crystals are up to 300 µm long, and equant to elongate, with aspect ratios up to 4:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, sector zoning is common, and some crystals appear to contain older cores. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 25–26 September 2020, using SHRIMP-B. Eleven analyses of the M257 standard obtained during the session indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.21% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Twenty-one analyses were obtained from 21 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to slightly discordant (Fig. 3), and can be divided into three groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios and their positions within the crystals.

Group I comprises 18 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 1781 ± 8 Ma (MSWD = 1.2).

Group X comprises two analyses of zircon cores (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates (1σ) of 1976 ± 14 and 1817 ± 9 Ma.

Group P comprises one analysis (Table 1), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 1715 ± 22 Ma (1σ).

Figure 1. **Hand specimen image for sample 225797: porphyritic microgranite, Three Peak Hill**

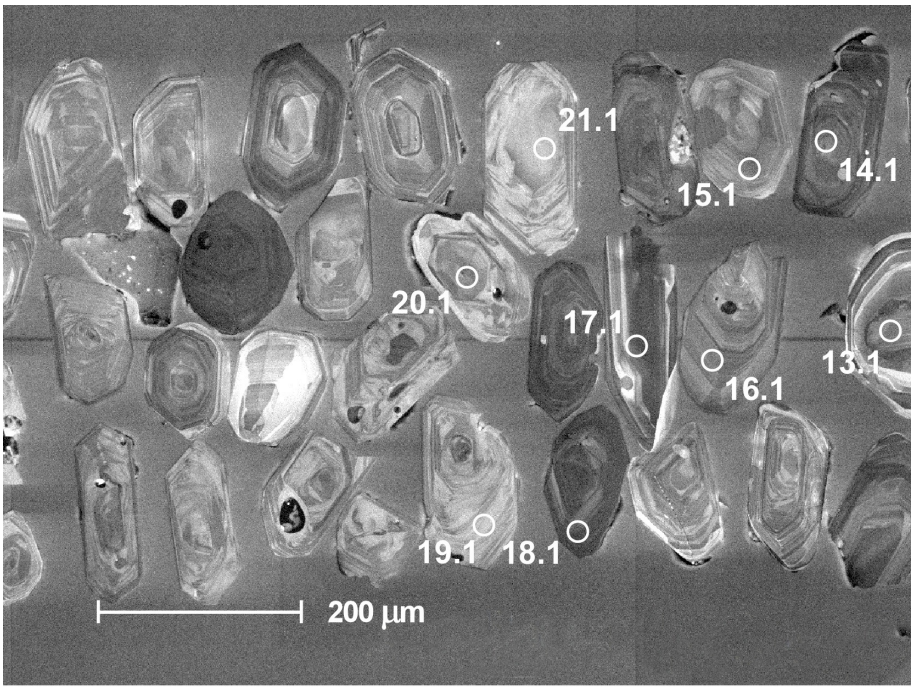

Figure 2. **Cathodoluminescence image of representative zircons from sample 225797: porphyritic microgranite, Three Peak Hill. Numbered circles indicate the approximate locations of analysis sites**

Table 1. Ion microprobe analytical results for zircons from sample 225797: porphyritic microgranite, Three Peak Hill

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	9	9.1	72	57	0.80	0.308	3.178	0.044	0.10912	0.00141	3.188	0.044	0.10644	0.00185	1759	21	1739	32	-1.1
I	17	17.1	92	56	0.61	0.129	3.217	0.040	0.10788	0.00116	3.221	0.040	0.10676	0.00133	1743	19	1745	23	0.1
I	20	20.1	180	223	1.24	0.226	3.040	0.031	0.10891	0.00089	3.047	0.031	0.10694	0.00111	1830	17	1748	19	-4.7
I	21	21.1	69	54	0.77	0.406	3.078	0.045	0.11087	0.00150	3.091	0.045	0.10733	0.00209	1807	23	1755	36	-3.0
I	3	3.1	99	117	1.19	-0.081	3.165	0.038	0.10732	0.00112	3.162	0.038	0.10802	0.00123	1771	19	1766	21	-0.3
I	14	14.1	213	176	0.83	0.343	3.155	0.030	0.11119	0.00085	3.166	0.031	0.10820	0.00114	1770	15	1769	19	0.0
I	8	8.1	233	51	0.22	0.062	3.130	0.028	0.10882	0.00071	3.132	0.028	0.10829	0.00076	1786	14	1771	13	-0.9
I	10	10.1	285	369	1.29	0.047	3.214	0.041	0.10875	0.00071	3.216	0.041	0.10834	0.00075	1745	19	1772	13	1.5
I	18	18.1	487	123	0.25	0.034	3.209	0.025	0.10881	0.00052	3.210	0.025	0.10852	0.00054	1748	12	1775	9	1.5
I	15	15.1	184	238	1.29	0.071	3.129	0.031	0.10926	0.00087	3.131	0.031	0.10864	0.00095	1787	16	1777	16	-0.6
I	4	4.1	707	452	0.64	-0.017	3.140	0.023	0.10896	0.00044	3.140	0.023	0.10911	0.00044	1783	11	1785	7	0.1
I	7	7.1	147	89	0.61	0.033	3.248	0.035	0.10950	0.00196	3.249	0.035	0.10922	0.00198	1730	16	1786	33	3.2
I	16	16.1	189	144	0.76	-0.045	3.159	0.031	0.10906	0.00086	3.158	0.031	0.10946	0.00090	1774	15	1790	15	0.9
I	6	6.1	498	224	0.45	0.009	3.102	0.024	0.10970	0.00054	3.103	0.024	0.10962	0.00055	1801	12	1793	9	-0.4
I	11	11.1	132	144	1.09	-0.032	3.205	0.035	0.10977	0.00102	3.204	0.035	0.11004	0.00105	1751	17	1800	17	2.7
I	12	12.1	134	121	0.90	-0.155	3.154	0.034	0.10918	0.00100	3.149	0.034	0.11053	0.00116	1778	17	1808	19	1.7
I	5	5.1	131	149	1.13	0.000	3.210	0.035	0.11067	0.00102	3.210	0.035	0.11067	0.00102	1748	17	1810	17	3.4
I	19	19.1	79	66	0.84	-0.336	3.122	0.042	0.10797	0.00134	3.111	0.042	0.11091	0.00179	1797	21	1814	29	1.0
X	1	1.1	466	40	0.09	-0.026	3.137	0.024	0.11081	0.00054	3.136	0.024	0.11104	0.00056	1784	12	1817	9	1.8
X	13	13.1	156	92	0.59	0.024	2.879	0.030	0.12154	0.00095	2.880	0.030	0.12133	0.00097	1921	17	1976	14	2.8
P	2	2.1	120	116	0.97	0.172	3.175	0.036	0.10652	0.00103	3.180	0.036	0.10502	0.00123	1763	17	1715	22	-2.8

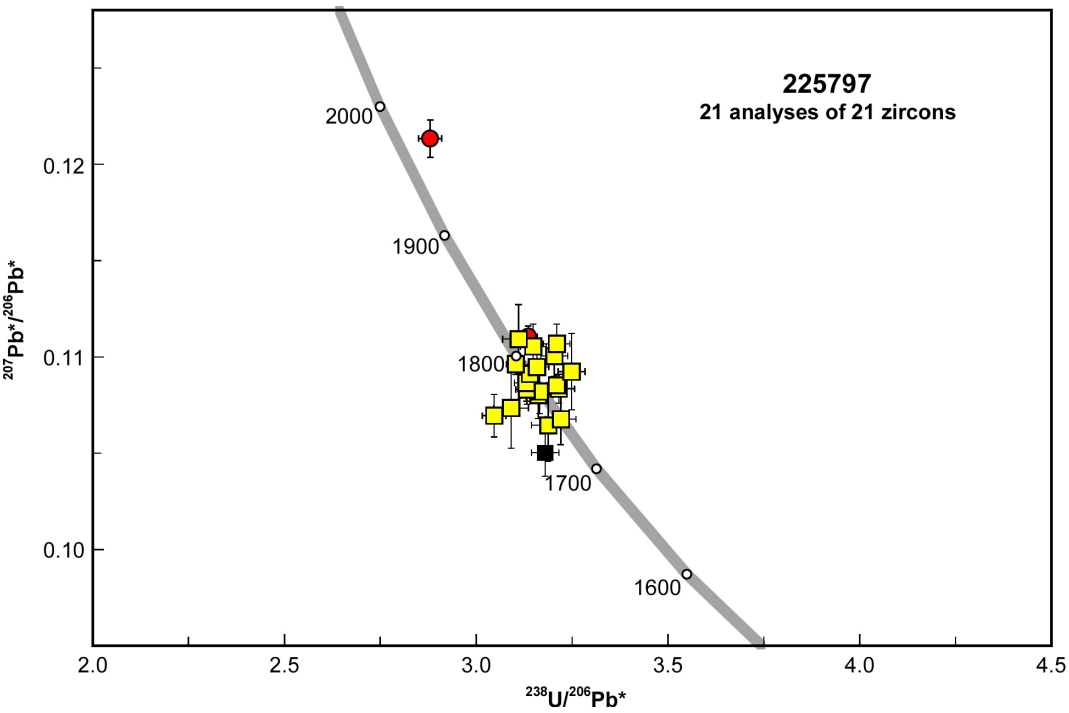

Figure 3. U–Pb analytical data for zircons from sample 225797: porphyritic microgranite, Three Peak Hill. Yellow squares indicate Group I (magmatic zircons); red circles indicate Group X (xenocrystic zircon cores); black square indicates Group P (radiogenic-Pb loss)

The date of 1781 ± 8 Ma for the 18 analyses in Group I is interpreted as the igneous crystallization age. The dates (1σ) of 1976 ± 14 and 1817 ± 9 Ma for the two analyses in Group X are interpreted as the ages of inherited zircons. The date of 1715 ± 22 Ma (1σ) for the single analysis in Group P is interpreted to reflect minor loss of radiogenic Pb.

References

Evans, DAD, Sircombe, KN, Wingate, MTD, Doyle, M, McCarthy, M, Pidgeon, RT and Van Niekerk, HS 2003, Revised geochronology of magmatism in the western Capricorn Orogen at 1805–1785 Ma: Diachroneity of the Pilbara–Yilgarn collision: Australian Journal of Earth Sciences, v. 50, p. 853–864.

Sheppard, S, Johnson, SP and Occhipinti, SA 2020, Moorarie Supersuite (P_-MO-g): Geological Survey of Western Australia, WA Geology Online, Explanatory Notes extract, viewed 07 September 2021, <www.dmp.wa.gov.au/ens>.

Stacey, JS and Kramers, JD 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221.

Thorne, AM and Johnson, SP 2020a, Ashburton Formation (P_-WYa-s): Geological Survey of Western Australia, WA Geology Online, Explanatory Notes extract, viewed 08 September 2021, <www.dmp.wa.gov.au/ens>.

Thorne, AM and Johnson, SP 2020b, Duck Creek Dolomite (P_-WYd-kd): Geological Survey of Western Australia, WA Geology Online, Explanatory Notes extract, viewed 03 September 2021, <www.dmp.wa.gov.au/ens>.

Thorne, AM and Seymour, DB 1991, Geology of the Ashburton Basin, Western Australia: Geological Survey of Western Australia, Bulletin 139, 141p.

Williams, IR, Ryan, GR and Halligan, R 1968, Yarraloola, WA Sheet SF 50-6: Geological Survey of Western Australia, 1:250 000 Geological Series.

Wingate, MTD, Lu, Y, Fielding, IOH and Johnson, SP 2019, 219588: porphyritic dacite, One Tree Well; Geochronology Record 1570: Geological Survey of Western Australia, 4p.

Wingate, MTD, Fielding, IOH, Lu, Y and Martin, DMcB 2021, 239798: porphyritic dyke, Pannawonica Road; Geochronology Record 1818: Geological Survey of Western Australia, 4p.

Recommended reference for this publication

Wingate, MTD, Fielding, IOH, Lu, Y and Martin, DMcB 2021, 225797: porphyritic microgranite, Three Peak Hill; Geochronology Record 1814: Geological Survey of Western Australia, 4p.

Data obtained: 26 September 2020

Data released: 12 November 2021